

DonorHyperglycemiaStatsProject

Running Code

Logistic Regression

```
m1 <- glm(delayed_fn ~ d_age + d_cod + d_caucasian + d_genderf + donorrisk,
          data = DonorHyperglycemia,
          family = "binomial")

summary(m1)
```

Call:

```
glm(formula = delayed_fn ~ d_age + d_cod + d_caucasian + d_genderf +
     donorrisk, family = "binomial", data = DonorHyperglycemia)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-2.106469	0.628056	-3.354	0.000797	***
d_age	-0.005237	0.008304	-0.631	0.528224	
d_cod2	-0.312951	0.301977	-1.036	0.300045	
d_cod3	-0.255460	0.324765	-0.787	0.431517	
d_cod4	0.763904	0.565541	1.351	0.176776	
d_caucasian	-0.013151	0.274557	-0.048	0.961797	
d_genderf	-0.094485	0.247659	-0.382	0.702822	
donorrisk	1.046880	0.403866	2.592	0.009538	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 497.37 on 429 degrees of freedom
Residual deviance: 482.50 on 422 degrees of freedom
(142 observations deleted due to missingness)
AIC: 498.5

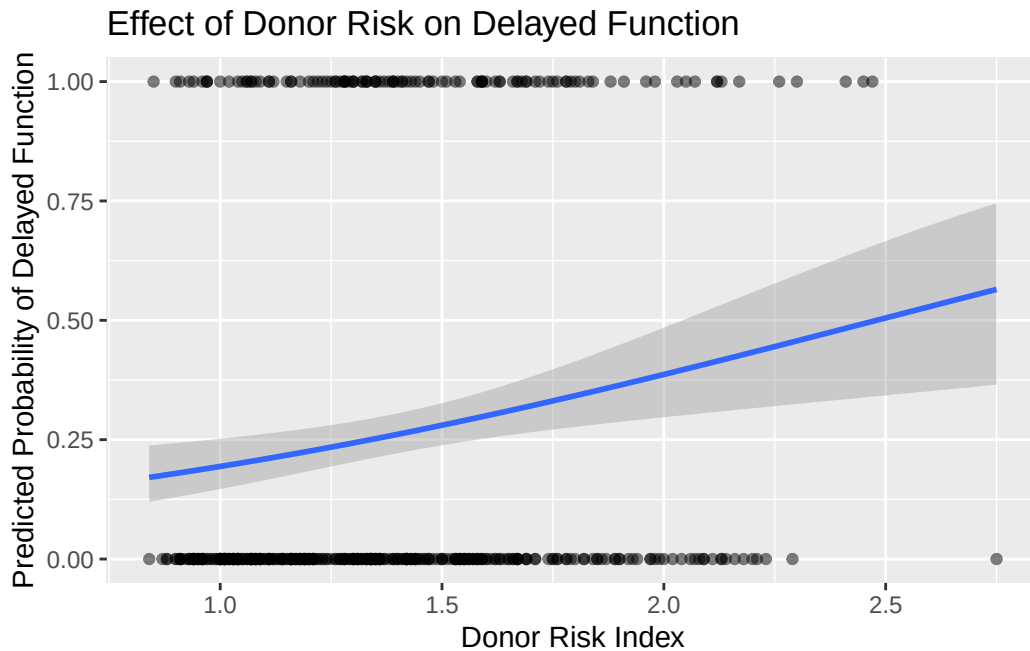
Number of Fisher Scoring iterations: 4

```
library(ggplot2)
ggplot(DonorHyperglycemia, aes(x = donorrisk, y = as.numeric(delayed_fn))) +
  geom_point(alpha = 0.5) +
  stat_smooth(method = "glm",
              method.args = list(family = "binomial"),
              se = TRUE) +
  labs(title = "Effect of Donor Risk on Delayed Function",
       x = "Donor Risk Index",
       y = "Predicted Probability of Delayed Function")
```

`geom\_smooth()` using formula = 'y ~ x'

Warning: Removed 135 rows containing non-finite outside the scale range
(`stat\_smooth()`).

Warning: Removed 135 rows containing missing values or values outside the scale range
(`geom\_point()`).

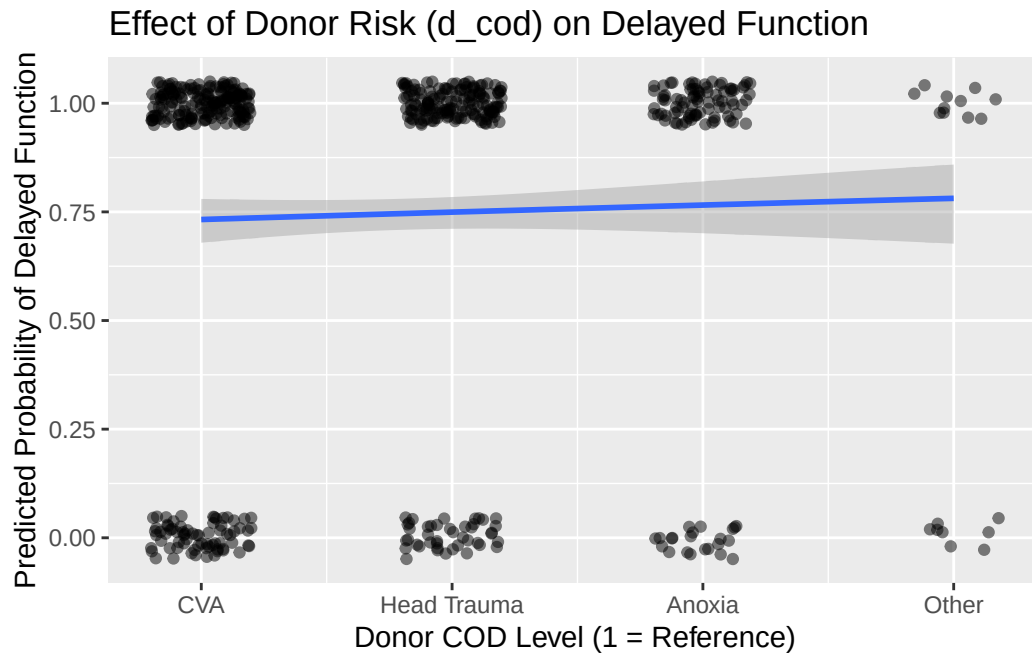


```
library(ggplot2)
ggplot(DonorHyperglycemia, aes(x = as.numeric(d_cod), y = as.numeric(delayed_fn != 1))) +
  geom_jitter(height = 0.05, width = 0.2, alpha = 0.5) +
  stat_smooth(method = "glm",
              method.args = list(family = "binomial"),
              se = TRUE) +
  scale_x_continuous(breaks = 1:4, labels = c("CVA" , "Head Trauma", "Anoxia" , "Other")) +
  labs(title = "Effect of Donor Risk (d_cod) on Delayed Function",
       x = "Donor COD Level (1 = Reference)",
       y = "Predicted Probability of Delayed Function")
```

`geom\_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat\_smooth()`).

Warning: Removed 1 row containing missing values or values outside the scale range
(`geom\_point()`).



```
library(ggplot2)
ggplot(DonorHyperglycemia, aes(x = d_cod, y = as.numeric(delayed_fn) - 1)) +
  geom_jitter(alpha = 0.5, height = 0.05, width = 0.2) +
  stat_smooth(aes(group = 1),
              method = "glm",
              method.args = list(family = "binomial"),
              se = TRUE) +

  labs(title = "Effect of Donor Risk on Delayed Function",
       x = "Cause of Death",
       y = "Predicted Probability of Delayed Function")
```

`geom\_smooth()` using formula = 'y ~ x'

Warning: Failed to fit group 1.

Caused by error:

! y values must be 0 <= y <= 1

